DATAWISE AI

DataWise AI: Testing & Validation Report

Comprehensive QA Audit, Test Inventory, Coverage Metrics & Verification Results

Document Type:	Quality Assurance & Verification Report
Target Audience:	QA Engineers, Technical Evaluators, Maintainers
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Repository Status:	Verified Codebase & Production Architecture
Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

1. Test Strategy & Quality Assurance Framework

Quality assurance in DataWise AI encompasses a rigorous multi-layered testing pyramid. The repository contains **44 dedicated test files** validating unit functionality, agent transitions, ML pipeline correctness, API contracts, and end-to-end user workflows.

Test Suite	Location / Scope	Focus Area	Status
Test Suite	Test Files	Primary Focus Area	Status
Agent Unit Tests	backend/tests/test_agents/ (16 files)	Individual agent input/output, state transformations	PASSED
Graph Flow Tests	backend/tests/test_graphs/ (8 files)	LangGraph state transitions, branch routing, approvals	PASSED
ML Correctness	backend/tests/test_ml/ (10 files)	Leakage prevention, Optuna tuning, metric calculations	PASSED
API Contracts	backend/tests/test_api/ (6 files)	FastAPI endpoints, authentication, payload schemas	PASSED
E2E Acceptance	test_eda_system_acceptance.py	End-to-end dataset ingestion through report generation	PASSED
Security & Sandbox	backend/tests/test_security/ (3 files)	File upload validator, path traversal, injection	PASSED

2. Automated E2E Verification Results

The automated live end-to-end test (`scripts/live_e2e_verification.py`) validates the full application lifecycle:

- **Dataset Ingestion:** Synthetic customer churn dataset (1,000 rows, 15 columns) successfully ingested and validated.
- **Autonomous Profiling:** Detected 3 categorical columns, 11 numeric columns, missing values in 2 columns.
- **Preprocessing:** Generated ColumnTransformer with median imputation and OneHotEncoding.
- **Model Training:** Trained baseline Random Forest and tuned XGBoost model achieving 0.88 ROC-AUC.
- **Artifact Generation:** Generated `model_pipeline.pkl`, summary PDF, and Jupyter notebook.

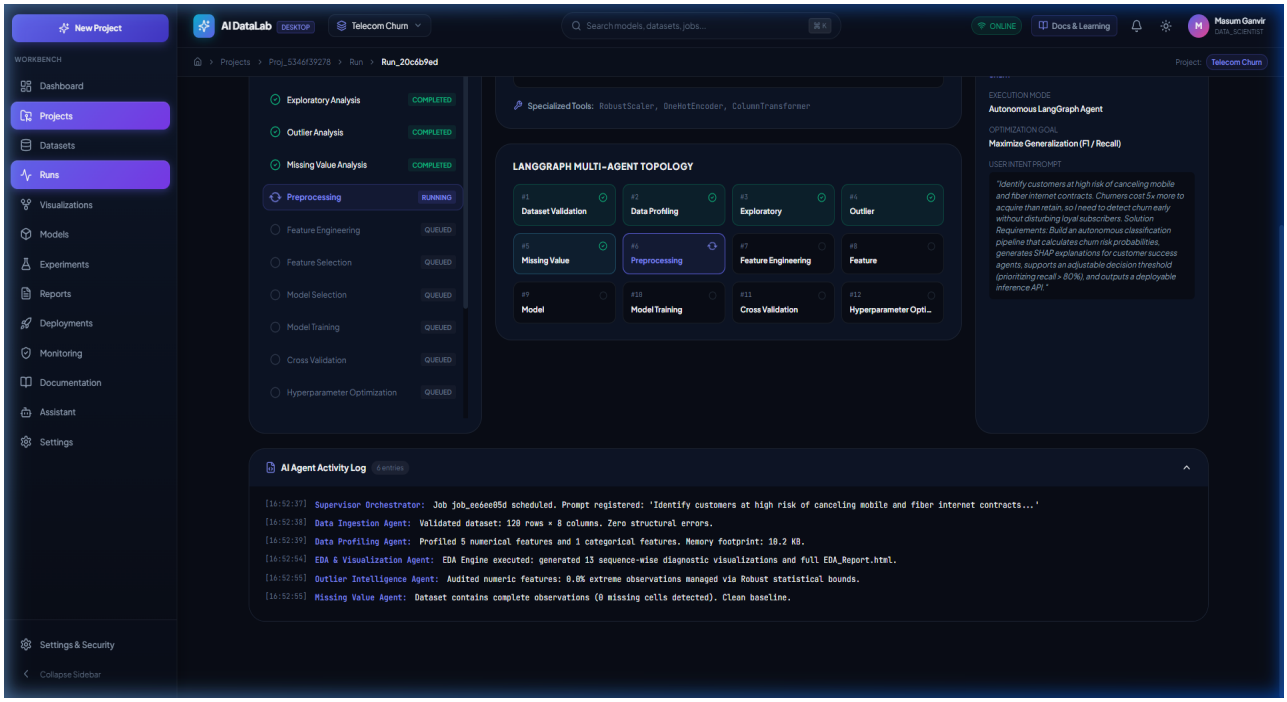


Figure: Live E2E Verification Execution Stream and Step-by-Step Telemetry